Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Rafael Gibrial	rafaelg@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Since I am a Computer Engineering major in the Cybersecurity Scholars Program, my goal is to build a Drafter website that serves as a password security analyzer that will evaluate the strength of the user's inputted password, calculating its information complexity, and estimating the time required to crack it via brute force attacks.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There is a website called How Secure is My Password? and other ones like it that analyze password complexity. Another one is called Password Strength Meter.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I would want this Drafter application to be used as a useful tool to inform people about their password strength while also making it an entertaining thing that is cool to play around with. I could implement a game mechanic to have people see who can get the longest time within

a set of constraints.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

It will calculate password strength, bit entropy, and brute-force crack time duration and output those values in a fun and engaging way.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

A text representing the user's password to be analyzed.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will evaluate some conditionals within the string to see if it contains lowercase, uppercase, symbols, special characters, and numbers for the passwords strength meter and then it will calculate the complexity of the password using a determined equation for its entropy.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

A breakdown of how string their password is, what it includes and doesn't include, and how long the password would take to crack by bruteforce.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Just the “Password”

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need: password length, presence of lowercase letters, presence of uppercase letters, presence of numbers, presence of special characters, and calculated entropy. Details I will ignore: The literal meaning of the password, user keyboard typing speed, and password reuse.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The part that seems the most complex is calculating of the password entropy and relating that value to the length of brute-force time to crack. Also, it seems kind of hard to showcase those results to the user in a format that will be easy to understand.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will use an LLM for conceptual guidance of the entropy formulas and then I will translate that to code. I will write the logic myself and manually test the calculations against passwords. The LLM will provide assistance on the “how to think about how to tackle this problem” part of this project.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

I want to have fun with this and showcase it to my friends because I feel like this would be a fun thing to play around with and just make it so good for the sake of making it good.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

Hey friend I want to build a password strength checker and brute-force time estimator website. What are 3 common problems that users face with password strength tools like this, and what are 3 simple features that this website should have?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

1. “I type out the password and then I forget what I wrote because I can’t see the password”
2. “I don’t believe that the password will last as long as it says it does”
3. “I like having my passwords short so those websites suggest passwords that are strong but too long”

SO they suggested:

1. “Having a way to see the password that i am typing in and saving it there”
2. “Showcasing the way it calculates the time estimation”
3. “Making a parameter that emphasizes shortness of the text while keeping it strong”

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

"Making a parameter that emphasizes shortness of the text while keeping it strong"
This was the most complex feature, and it is complicated because this website is meant to analyze and not really suggest passwords for the user to use.

Simpler alternative:
"So then you can just focus exclusively on the evaluation of the password without the need to also have password generation in the website"

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

Based on the conversation, the project scope shifted from a potential password generator and analyzer to just a password evaluation of whatever the user types in. This will then display the mathematics behind the time-to-crack estimation and will include the toggle to view the password.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will use Drafter documentation, Python math documentation, and my friend's advice on the user interface and web mechanics.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

1. I will check Drafter/Bakery error log and locate the bug.
2. I will check what type of bug it is and test it out by its lonesome
3. Check the inputs and the returns of the part of code
4. And consult the drafter documentation to relieve the issue

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

For this assignment, I didn't use an LLM other than a google search on password complexity.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.